

Trading Idea: LLM-Based Football Match Prediction for World Cup Betting Markets

Logic

The core idea is simple: sports betting markets for World Cup matches are driven heavily by narrative. Public bettors and even bookmakers overweight team reputation, recent media coverage, and national sentiment when pricing match odds. A model that instead reasons from granular, player-level statistical data across multiple leagues and tournaments should be able to produce more calibrated predictions, especially for matches where the narrative diverges from the underlying team composition.

I fine-tuned a LLaMA 3.1 8B language model using QLoRA on historical football data. The training set covers every player from World Cup qualifying nations, including their individual statistics from club leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1, etc.) and past international tournaments. For each historical match, the model receives a structured prompt containing both teams' full rosters with per-player stats (goals per 90, assists, pass completion, defensive actions, form trajectory), and learns to predict the match outcome along with a natural-language reasoning chain explaining its prediction.

Pattern

This strategy sits at the intersection of statistical and behavioral patterns. On the statistical side, the model captures complex relationships between player-level features and match outcomes that would be difficult to hand-engineer into a traditional regression or tree-based model. On the behavioral side, it exploits the tendency of betting markets to misprice World Cup matches due to narrative bias. World Cup matches are particularly susceptible because they happen infrequently, involve national pride, and attract large volumes of casual bettors whose decisions are sentiment-driven rather than data-driven.

Model and Trading Rules

The signal is the model's predicted probability for each match outcome (win, draw, loss), accompanied by a reasoning chain that explains the prediction by referencing specific player statistics. Before acting on any signal, the reasoning is reviewed for logical soundness. If the model cites a player who is injured or suspended, the signal is overridden. Only predictions where the model's confidence exceeds a minimum threshold are considered actionable. Position sizing follows fractional Kelly criterion based on the magnitude of the predicted edge. A hard cap limits maximum exposure per match, and a drawdown circuit breaker pauses all activity if cumulative losses exceed a predefined percentage of the bankroll.

Expected Outcomes

The model will be backtested against the 2022 FIFA World Cup (64 matches) as ground truth. Key metrics include prediction accuracy (win/draw/loss), score prediction proximity, and reasoning quality. The 2022 World Cup is a strong test case because it featured several major upsets (Saudi Arabia over Argentina, Japan over Germany) where narrative-driven models would have failed. A secondary backtest on 2022 league matches is planned to validate generalization. The goal is not necessarily to beat a professional bookmaker, but to demonstrate that a fine-tuned LLM can produce well-calibrated, interpretable match predictions grounded in player-level evidence, and that this approach can be structured as a disciplined, rule-based trading strategy.